PAOLO VIVIANI

Research Engineer

June 14th, 1989 | paolo.vivi@gmail.com | +39 329 1865014 | Torino, ITALY | paoloviviani.github.io

EDUCATION

B.Sc. in Physics

University of Torino, 104/110

2011 | Collegio Universitario “R. Einaudi”, Torino

Scholarship winner: "Piano Lauree Scientifiche", granted by Società Italiana di Fisica.

M.Sc. in Theoretical physics

University of Torino, 104/110

2015 | Collegio Universitario “R. Einaudi”, Torino

Ph.D. in Computer Science

University of Torino, Computer Science Dept.

2015 – 2019 | Torino, IT

Supervisor: Marco Aldinucci. Funded by Noesis Solutions.

EXPERIENCE

Noesis Solutions

Reasearch Engineer

April 2015 – Ongoing | Novara, IT

- Development of machine/deep learning methodologies for engineering.
- Re-definition of the internal development and deployment stack (C++, Python) for numerical computations, with focus on performance and maintainability.
- Cloud and container technologies for the deployment of engineering workflows.
- Reviewed whole internal source code management workflow.
- Technical responsible for a few research projects. Including presentation to European Commission reviewers.

Research Projects

MACH

ITEA2 project

April 2015 – December 2016 | www.mach-project.org

Fortissimo 2

Horizon 2020-FoF project

February 2016 – Ongoing | www.fortissimo-project.eu

CloudFlow

FP7-I4MS project

March 2016 – May 2017 | www.eu-cloudflow.eu

Blockchain for online Service Security (BoSS)

IMEC-ICON flemish project

March 2018 – ongoing

FIELDS OF INTEREST

- High Performance Computing
- Systems for Deep learning
- Containers and Cloud deployment

SKILLS

Parallel Computing	●●●●●
GPU Computing	●●●○○
Cloud technologies	●●●●○
Machine Learning	●●●●○

Programming

C++	●●●●●
Python	●●●●○
MPI/OpenMP and other tools	●●●●○
CUDA	●●○○○
Git	●●●●○

You won’t find any Java here, please desist.

Tools

MxNet	●●●●●
Keras	●●●○○
Apache Spark	●●○○○
Numerical Libraries	●●●●○
Linux OS	●●●●○
Openstack	●●●○○
Docker	●●●●○

CISCO CCNA course completed in 2008.

Soft

Presentations	●●●●○
Data visualization	●●●○○
Factorization of problems	●●●●●
Time management	●○○○○
Workarounds	●●●●●

LANGUAGES

Italian	●●●●●
English	●●●●●
French	●●○○○

ACADEMIC

Publications

- Reniers, Vincent et al. “Analysis of Architectural Variants for Auditable Blockchain-based Private Data Sharing”. In: *In proc. of the 34th ACM Symposium on Applied Computing (SAC)*. Limassol, Cyprus, Apr. 2019.
- Viviani, Paolo, Maurizio Drocco, Marco Aldinucci, and Marco Grangetto. “Accelerating spectral graph analysis through wavefronts of linear algebra operations”. In: *Proc. of 27th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)*. Pavia, Italy: IEEE, 2019.
- Viviani, Paolo, Maurizio Drocco, Daniele Baccega, and Marco Aldinucci. “Deep Learning at Scale”. In: *Proc. of 27th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)*. Pavia, Italy: IEEE, 2019.
- Aldinucci, Marco et al. “HPC4AI, an AI-on-demand federated platform endeavour”. In: *ACM Computing Frontiers*. Ischia, Italy, May 2018.
- Tordini, Fabio, Marco Aldinucci, Paolo Viviani, Ivan Merelli, and Pietro Liò. “Scientific Workflows on Clouds with Heterogeneous and Preemptible Instances”. In: *Proc. of the Intl. Conference on Parallel Computing, ParCo 2017, 12-15 September 2017, Bologna, Italy*. Advances in Parallel Computing. IOS Press, 2018.
- Viviani, P., M. Aldinucci, R. d’Ippolito, J. Lemeire, and D. Vucinic. “A Flexible Numerical Framework for Engineering—A Response Surface Modelling Application”. In: *Improved Performance of Materials: Design and Experimental Approaches*. Cham: Springer International Publishing, 2018, pp. 93–106.
- Viviani, Paolo, Maurizio Drocco, and Marco Aldinucci. “Pushing the boundaries of parallel Deep Learning - A practical approach”. In: *CoRR* abs/1806.09528 (2018).
- – “Scaling Dense Linear Algebra on Multicore and Beyond: a Survey”. In: *Proc. of 26th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)*. Cambridge, United Kingdom: IEEE, 2018.
- Viviani, Paolo, Massimo Torquati, Marco Aldinucci, and Roberto d’Ippolito. “Multiple back-end support for the Armadillo linear algebra interface”. In: *In proc. of the 32nd ACM Symposium on Applied Computing (SAC)*. Marrakesh, Morocco, Apr. 2017, pp. 1566–1573.
- Viviani, Paolo, Marco Aldinucci, and Roberto d’Ippolito. “An hybrid linear algebra framework for engineering”. In: *Advanced Computer Architecture and Compilation for High-Performance and Embedded Systems (ACACES) – Poster Abstracts*. Fiuggi, Italy, July 2016.
- Viviani, Paolo, Marco Aldinucci, Roberto d’Ippolito, Jean Lemeire, and Dean Vucinic. “A flexible numerical framework for engineering - a Response Surface Modelling application”. In: *10th Intl. Conference on Advanced Computational Engineering and Experimenting (ACE-X)*. 2016.

Other

- Program Committee member, *16th IEEE International Conference on Scalable Computing and Communications (ScalCom 2016)*
- Program Committee member and session chair, *Euromicro International Conference on Parallel, Distributed, and Network-based Processing (PDP2018)*
- Program Committee member, *Parallel Numerical Methods and Libraries for Heterogeneous Multi/Manycores (Special Session of PDP2018)*
- Program Committee member, *Artifact Evaluation, Euro-Par 2018 Conference*
- Program Committee member, *Euromicro International Conference on Paralle, Distributed, and Network-based Processing (PDP2019)*
- Program Committee member, *Parallel Numerical Methods and Libraries for Heterogeneous Multi/Manycores (Special Session of PDP2019)*

FREE TIME

Avid traveller and photographer: 4 continents, 21 countries visited, and more to come.
Former alleged athlete, now turned to playground basketball player.

`paoloph.carbonmade.com`

Photographer

OASIS Project

2013 Kharga, Egypt

Archaeological expedition funded by Fondazione Collegio delle Università Milanesi and American University in Cairo.

Photographer

Collegio Universitario "R. Einaudi"

2012 – 2014 Torino, Italy

Exchange student

Tauranga Boy’s College

2006 Tauranga, New Zealand

Scuba Diver

PADI Advanced Open Water Diver

From 2004

Track and Field athlete

Javelin Throw – Regional and National competitions

1997 – 2012

Volunteer

IAAF Athletics World Championship

2009 Berlin, Germany

English Course

‘General English Super Intensive’ C1 level course at Alpha College of English

2012 Dublin, Ireland

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